

Teorema de diferenciación de Lebesgue en $\mathcal{L}^1$

Teorema 1 (de diferenciación de Lebesgue en $\mathcal{L}^1$). Sea $f \in \mathcal{L}^1(\mathbb{R})$, entonces

$$\lim_{h \rightarrow 0} \frac{1}{2h} \int_{x-h}^{x+h} f(t) dt = f(x) \quad \text{c.t.p. } x \in \mathbb{R}.$$

Demostración: Por el lem-diferenciacion-lebesgue-l1-imp-casi-toda-parte/??, tenemos que $\forall n \in \mathbb{N}$:

$$A_{1/n} := \left\{ x \in \mathbb{R} : \limsup_{r \rightarrow 0^+} \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y) dy - f(x) \right| > \frac{1}{n} \right\} \text{ tiene medida de Lebesgue cero.}$$

Tomamos $A = \bigcup_{n=1}^{\infty} A_{1/n}$, entonces $m(A) = 0$. Entonces, para $x \in A^c$, tenemos que $\forall n \in \mathbb{N} : x \notin A_{1/n}$, es decir,

$$\forall n \in \mathbb{N} : \limsup_{r \rightarrow 0^+} \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y) dy - f(x) \right| \leq \frac{1}{n} \implies \lim_{r \rightarrow 0^+} \frac{1}{2r} \int_{x-r}^{x+r} f(y) dy = f(x). \quad \blacksquare$$

Referenciado en

- teo-diferenciacion-lebesgue

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